

Special Issue on Emerging and Renewable Energy: Generation and Automation

GIS-Based Method for Future Prospect of Energy Supply in Algerian Road Transport Sector Using Solar Roads Technology

Belkhir Settou^{*}, Nouredine Settou, Abderrahmane Gouareh, Belkhir Negrou,
Charafeddine Mokhtara, Djilali Messaoudi

VPRS laboratory, University of Ouargla, BP 511 Ouargla 30000. Algeria

Abstract

This paper aims to investigate the possibility of integration of Electric Vehicles EVs supply's with electricity and/or hydrogen in the road transport sector and estimate the energy supply derived from solar irradiation by using solar roads technology. The case study is road Est-Oust (road E-O) of Algeria. A Geographic Information System and spatial analysis tools are combined with spatial data and technical models to carry out these calculations. The results of this study demonstrate that solar road panels, which are integrated into the road E-O, produce over to 804 GWh/year which equivalents to 13778 tons of hydrogen per year. by using FCEVs will saving over then 41.10^3 liter of fossil fuels (regular gasoline); and reduce GHG emission (CO_2) in the transportation sector by 216 tons per year.

© 2019 The Authors. Published by Elsevier Ltd

Selection and peer-review under responsibility of the scientific committee of the 6th International Conference on Emerging and Renewable Energy: Generation and Automation, ICEREGA 2018.

keywords: Solar road panel; Climate change; GIS; Solar energy; Transport sector; Electric vehicle; Hydrogen; GHG emission

1. Introduction

In the modern world, the transportation sector based mainly relies on fossil fuel. Using large amount of fossil fuel is responsible for global warming, ozone layer depletion and air pollution. And is responsible for about a 45% of total

^{*}Corresponding author: +213-665-808-808

E-mail address: belkhir.settou@gmail.com

final energy consumption [1]. According to Energy Information Administration (EIA) the transportation sector occupies almost 40 % of the world's total energy consumption and 29% Carbon dioxide gas emission in 2015 [2].

The absence of any alternative solution for transportation sector; which makes this sector keep going to that trend. As an alternative to fossil fuels, the promising way of doing this is to replace the traditional vehicles, with their internal combustion engines (ICE), with electric vehicles (EVs), which are one of the viable solutions for saving the environment and decreasing the GHG emission.

There are many factors influencing GHG emissions from road transportation sector such as the increasing population growth, the increasing number of automobiles or high dependency on cars, limited public transportation service facilities and cheap fuel (gasoline/ diesel) [3], [4].

The growing emissions of these fuels from the transport sector may significantly accelerate the future deployment of hydrogen as many literatures [5], [6], a detailed overview of diverse hydrogen production technologies along with environmental and safety aspects of its use as an energy carrier are reported by [7]. A comparative review of hydrogen's role in low carbo energy over then other technologies are presented by [8], and they have found that EVs are the method chosen for decarbonization in the transportation sector. Five scenarios of sales and GHG emission factor are studied by [9] according to their results, The GHG emissions will decrease in "Fuel Cell vehicles" scenario (which is integration an 8 million FCVs in 2050) by 13.9% compared with "no-FCVs" scenario ,which explain by [10], [11] they studies the possibility of integration the EVs in roads transportation sector. This paper aimed to investigating the possibility of integration of EVs fueled with electricity and/or hydrogen in the road transport sector and estimate the energy supply derived from solar irradiation by using solar roads technology. The case study is road Est-Oust (road E-O) of Algeria. A Geographic Information System (GIS) and spatial analysis tools are combined with spatial data and technical models to affecting these calculations.

Nomenclature

BEVs	Battery Electric Vehicles
E_{H_2}	Hydrogen energy produced
E_{SRP}	Electrical energy form solar road panel
EVs	Electric Vehicles
FCEVs	Fuel Cell Electric Vehicles
G	Annual horizontal solar irradiation
GIS	Geographic Information System
HHV_{H_2}	Hydrogen Higher heating value
ICE	Internal Combustion Engines
L	Distance travelled
M_{H_2}	Annual hydrogen production
η_{Elec}	Electrolyzer operation efficiency
η_{SRP}	Solar Road Panel efficiency

1.1 Overview of the Algerian energetic context

Algeria is the largest area in the Arabian and African region and the geographical location in North Africa gives it a very important strategic position on the African and international level. With an area of 2,381,741 km² Algeria is the tenth largest country in the world. It is bordered to the north by the Mediterranean and Tunisia by the north-east. In the east with Libya, to the south are Niger, Mali and Mauritania, in the south-west with the Western Sahara, bordered by Morocco to the west. The heavy dependence of the economy on the oil and gas sector is a potential source of vulnerability for the Algerian economy. The oil and gas alone provide more than a third of Gross Domestic Product (GDP) (36% in 2013)[12].

Algeria is the seventh largest country in the world in terms of proven natural gas reserves of up to 4 Trillion cubic meters. While the value of proven oil reserves is 12 billion barrels at the end of 2016 [13]. In MENA (Middle East

and North Africa) region. Algeria is one of the most petroleum exporting countries, especially natural gas worth 54 billion cubic meters[2]. Europe is the traditional market for Algeria's hydrocarbon exports, with 72 % of exports destined for it. And in the second place USA with 16%. According to [13].Algeria's natural gas exports to Italy and Spain via pipelines are estimated at 17.2 billion cubic meters and 11.8 billion cubic meters respectively. And via tankers to Europe and Eurasia in the form of liquefied natural gas (LNG) with total value 14.9 billion m3 which is 93% from total exportation Algerian of LNG. The consumption of electricity present 29.1 % of final energy in Algeria which is in the 3th place after oil and NG. At the sectoral level; the residential sector is a substantial consumer of energy by 43 % followed by the transport sector by 35.1% [12].By the end of 2018, 93% of transportation fuels are derived from petroleum product ;Which includes: Gasoline, Diesel, LPG; the amount of gasoline and diesel consumed in Algeria had decreased at an annual average rate of 2.6% compared to 2016, which is due to the increase in fuel prices by 35% in the Gasoline and Diesel but the stability of the price of gas led to an increase in consumption more by 21% in this period. That's will appear more in the next two years because the government has raised support for prices again [12], [14].

1.2 Overview of electric vehicle technology

Rapidly developing power electronics technologies has even realized high energy-efficient vehicles. Those vehicles are in three family, namely internal combustion engine vehicle (ICEV), which based on internal combustion engine, hybrid electric vehicle (HEV) is hybridization configuration of (ICEV) and all electric vehicle (AEV) where they use only electricity as energy source.

All electric vehicles or AEVs are vehicles based on used electrical power as source for vehicle propulsion According to [15] there are 6 types of power transfer configuration depending on the energy source this type of vehicles can be into three major types Battery Electric Vehicles (BEVs), Fuel Cell Electric Vehicles (FCEVs) and Fuel Cell Hybrid Electric Vehicles (FCHEVs).

The Fig.1 present combination design of Fuel Cell and battery Electric Vehicles (Plug-in-FCEVs). The main powertrain components in a typical fuel cell vehicle has listed as; fuel cell stack, high-pressure hydrogen supply system or reformer with a fuel tank, battery, DC/ DC conversion ,DC/AC conversion and traction motor (includes a transmission). The battery can be used as Energy Storage System (ESS) as it can be charged and discharged based on the power demand and supply. Power conversion in this configuration of Plug-in-FCEVs has two stages; DC-DC conversion and DC-AC conversion. First one which allow to: convert a low voltage DC power from ESS or FC into high voltage DC and it provides a stable voltage level on DC bus and restricts large voltage swings. The second one transfer's power from the input DC bus of traction motor to AC power supplied; which is produces by FC and ESS, the wheel of the vehicle uses a mechanical output from traction motor (AC power) to drive it.

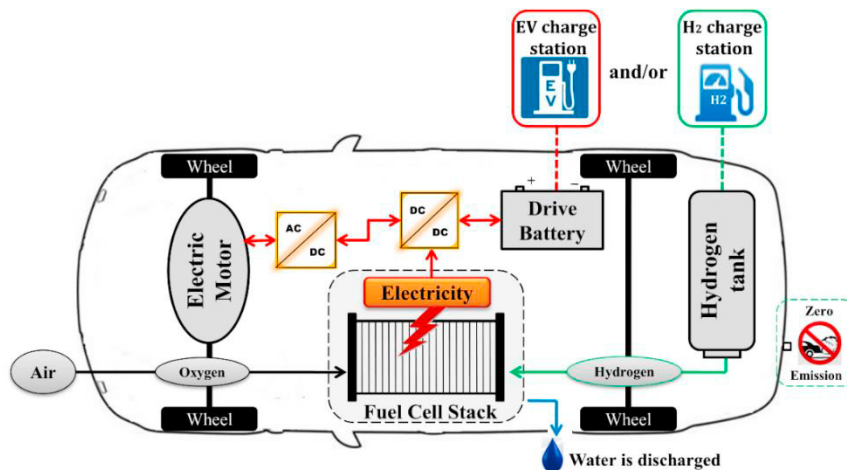


Fig. 1: Plug-in-FCEVs configuration (Adopted and re-figured from Ref [16])

1.3 Solar road system design and energy generation uses

An advanced penetration of Plug-in-FCEVs in transport sector is essentially focused on advancing electricity and/or hydrogen fueling infrastructure and charging stations to support more energy supply. The availability of stations providing electricity and/or hydrogen in places where Plug-in-FCEVs will be deployed remains a key challenge to the adoption of this kind of vehicle. Solar road technology is suggested to address this challenge. It can play significant roles in creating the decentralized energy supply through charging station. Solar roads technology is a solar panel that could potentially be installed in the road to convert solar energy directly into electricity using semiconductor. The idea is to replace all current asphalt roads with solar panels that collect and store solar energy. This renewable energy is seen as a means of reducing dependence on fossil fuels as an electrical product. Each individual panel may consist of at least three sub layers: surface layers, electronics layer and base plate layer [17], as shown in **Fig.2**; the solar panel generate electricity directly from sunlight via an electronic process that occurs naturally in certain types of material, called semiconductors, this electricity produced will be used to supply power to charging stations of Plug-in-FCEVs. It's also designed to produce a hydrogen by broken down the water into its component elements by passing electricity through it. This process is called electrolysis. The hydrogen produced will be stored into tanks which guarantees a continuous supply of hydrogen to hydrogen charging stations.

This idea was realized in the first time in the world in 2014 by TNO laboratory in Noord-holland is province in the north west of the Netherlands. By a 70-meter section of a cycle path that generates electricity from solar panels; the TNO solarRoad project team assembled a team composed of many engineers in different fields: road construction, material science, solar energy, grid technology; All of them in order to realize the idea on the ground. [18]

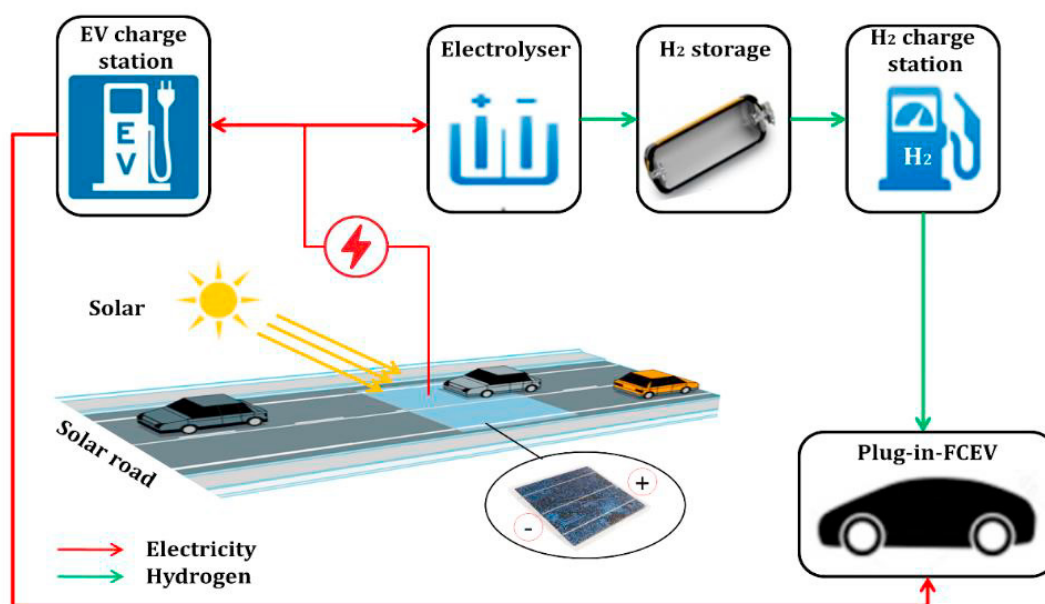


Fig 2: Schematic diagram of Plug-in-FCEVs energy supply based on solar roads technology.

In 2016, French company BTP COLAS has developed a new solar road panels (named Watt-Ways) for sections ranging from 10 m² to 50 m² [19]. Their size can be adapted to fit any type of road infrastructure (roads, highways, parking lots, bike paths, etc.). It can be installed directly on existing pavement without any need for civil engineering work [20]. Watt-way panel is polycrystalline silicon. According to [21] the estimation of electricity will reach an average of 280 MWh/year, and 1500 KWh/day in summer. China has opened also a km includes two lanes and an emergency lane of solar roads in Jinan; the capital of Shandong province South of Beijing; in 30th December 2017. The intents Qilu Transportation Development Group, is the responsible of developing this project. According to [22] this road can generate up to a million Kilowatt-hours of power annually. Enough to power 800 Chinese homes.

2. Methodology to assess the available solar road energy supply; case study road E-O

In order to evaluate the electricity and hydrogen production derived by solar roads technology for the road E-O of Algeria based on GIS modeling there is three important of calculation steps as shown in **Fig.3**. Two types of basic data are used for our calculations. One type of data consists of the attributes of roads which include the road names, road lengths, the locations (coordinates of each segment) and the traffic volume. The other type of data is for solar energy information which includes the yearly solar irradiation for each grid (pixel) given by (KWh/m². year).

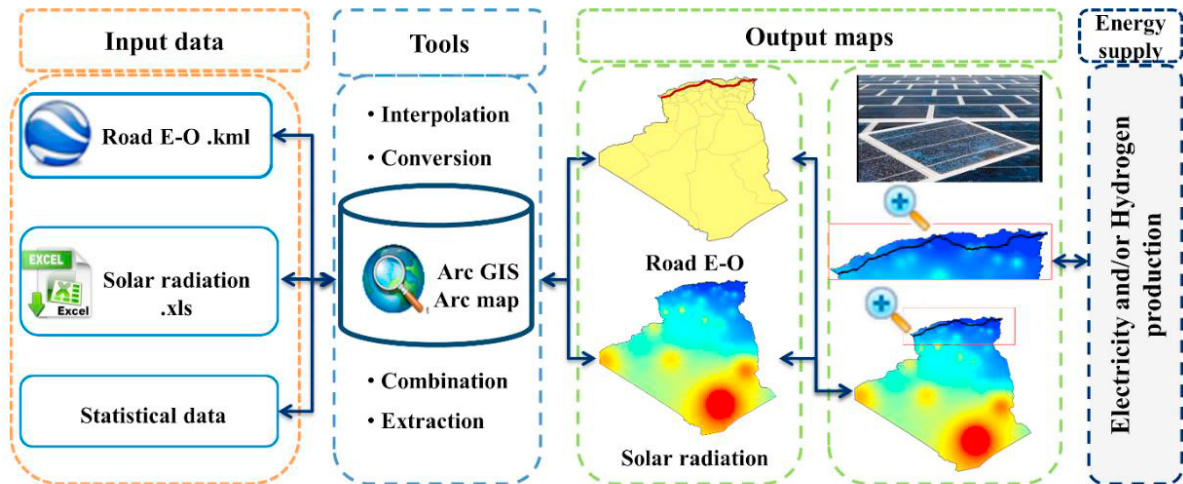


Fig.3 Schematic diagram to assess the available solar roads energy supply in the road E-O.

2.1 Data sources and preparations

2.1.1 Solar resource analysis

A variety of methods are used to assess and analyses the potential of solar energy in Algeria. In this section the important steps in the data collection & theirs preparation of the potential of solar energy in Algeria are described, which is varied due to the geographic site specifications, climatic conditions and technological limitations. The interpolation method was used to develop solar irradiation map from more than 40 solar data base points located at the cities center, this method is a tool available in geographical information system (ArcGIS 10.2) [23]. In this analysis, once of these methods the Inverse Distance Weighted (IDW) method is used to interpolate and determinate the available solar potentials in the whole territory of Algeria by using limited number of points [24]. As the **Fig.3** shown the solar irradiation map; an important potential of solar energy is available and remarkably in Algeria.

2.1.2 Digitization of the road Est-Oust

In order to generated road E-O polyline dataset. Google Earth, as the visualization platform and developed conversion tools (called 'KML generators') was used. Google Earth is geographical information program released in 2005 and present an actual visualization map of the globe. High-resolution satellite image available in Google Earth was sufficiently used for visualization and digitization of the road E-O. By using Google earth tools, the road E-O line was generated and stored as kml files (.kml) and then imported to ArcGIS 10.2 to be converted into shape file (.shp) which combined with Administrative division shape file which is presented by [25].

2.2 Yearly solar irradiation available in Road E-O:

The solar map is raster data type, which comprises a matrix of pixels (or cells), each containing a value that represents the solar irradiation for the area covered by that cell. Its size depended to the raster resolution, where resolution increases as the size of the cell decreases. The road E-O polyline is decomposed into over 12200 segments, where solar irradiation value for each segment is extracted using the cell covering its area. Finally,

3. Estimation of solar electricity/hydrogen production

Solar road panels generate electricity directly from sunlight via an electronic process that occurs naturally in certain types of material, called semiconductors. Electrons in these materials are freed by solar energy and can be induced to travel through an electrical circuit, powering electrical device [26]. The annual energy produced by the solar road panel can be expressed as follows[27]:

$$E_{SRP} = \eta_{SRP} \times G \quad (1)$$

Where G the Annual horizontal solar irradiation [KWh/m²/year] and η_{SRP} Solar Road Panel efficiency, [%] To estimate the electrical energy potential, it was considered COLAS panels (described above) with an efficiency of 15% [19]. The electrical energy which can be produced from E-O solar roads is presented in **Fig.4**.

The electrolyzer driving the electrolysis process of water by solar generated energy. The system under study is considered produce renewable hydrogen based on use the proton exchange membrane electrolyzer (PEM); the PEM electrolyzer has an efficiency 75%. Furthermore, the rapid start-up and response as advantages[28]; which allow maximum generation of hydrogen during sunny hours in the day. This study set out to determine the electricity energy used by the proton exchange membrane electrolyzer (PEM). For this, on consider the electrolyzer consumed 53.2 KWh to produce one kg of hydrogen (which equals about 75% of efficiency) [27]; Power needed. Which should be sent to the electrolyte electrodes are defined as follows [29]:

$$E_{H_2} = \eta_{Elec} \times E_{SRP} \quad (2)$$

Where η_{Elec} [%] represent the Electrolyzer operation efficiency, E_{SRP} is Electrical energy form solar road panel [KWh/km²/year]. When the potential for the annual mean of the available solar electric energy by solar road panel is estimated, the annual hydrogen production from this renewable resource can be computation as follows [30]:

$$M_{H_2} = \frac{E_{H_2}}{HHV_{H_2}} = \frac{\eta_{Elec} E_{SRP}}{HHV_{H_2}} \quad (3)$$

Where M_{H_2} is the annual hydrogen production [kg/Km²/year], E_{H_2} is the hydrogen energy produced [KWh/km²/year] and HHV_{H_2} : the hydrogen higher heating value (39.4 KWh/kg).

3.1 Estimation CO2 Emission from road E-O

The greenhouse gas (GHG) emissions from the transport sector are considered severe, and the increasing of the fuel consumption due to the fast growing vehicles density on roads, which are the largest source for the major atmospheric pollutants. The CO₂ generated from burning one kilogram of fuel depends on the amount of carbon in the fuel. For example the carbon dioxide emitted by combustion 1 kg of gasoline and diesel is 3216.02 g and 3232.3 g respectively[31]. According to [32] for an average passenger vehicle. The annual CO₂ emission is calculated by:

$$\text{Annual } CO_2 \text{ emission (metric ton)} = \frac{CO_2 \text{ emission} \left[\frac{g}{kg} \text{ of gasoline} \right]}{\text{fuel economy} \left[\frac{km}{kg} \text{ of gasoline} \right]} \times L \quad (4)$$

Where L is distance travelled by car in km/year. A typical passenger vehicle with an annual drive over than 100,000 km (considering driving back and forth once a day in Road E-O). an estimation of fuel saving (Gasoline/Diesel) a comparative analysis between an average ICE and FCEV it was in two parts; an overview of the most widespread internal combustion engine vehicles in Algerian roads; which is based on statistics of National Office of Statistics [33] and the technical sheets of every type of vehicle as sources; in our comparative analysis of the selected vehicles which based on technical parameters (tanks volume, fuel consumption for every 100 km) that serve as criteria as an average vehicle in Algerian roads is :4.9 L per 100 km; which allow to compare it with FCEVs (Toyota Mirai 1 kg H₂ for each 100 km[34]).

4. Results and Discussion

Following the method described in the second section (Fig.3); and by using Eq. (1) and the values are presented in Fig.4, that will give an estimate of the electricity produced from RSP over than 804 GWh per year, the estimation of the solar irradiation available for each province is calculated based on yearly solar irradiation available and the area of segments, due to the unevenly distributed of irradiation solar among various regions of road E-O, which will be effect on electricity produced or hydrogen from this road; that's observed in case Skikda and Chlef provinces, they produced an average of electricity close then 50 GWh/yr but and difference of nearly 3 km in segment length of road E-O between them.

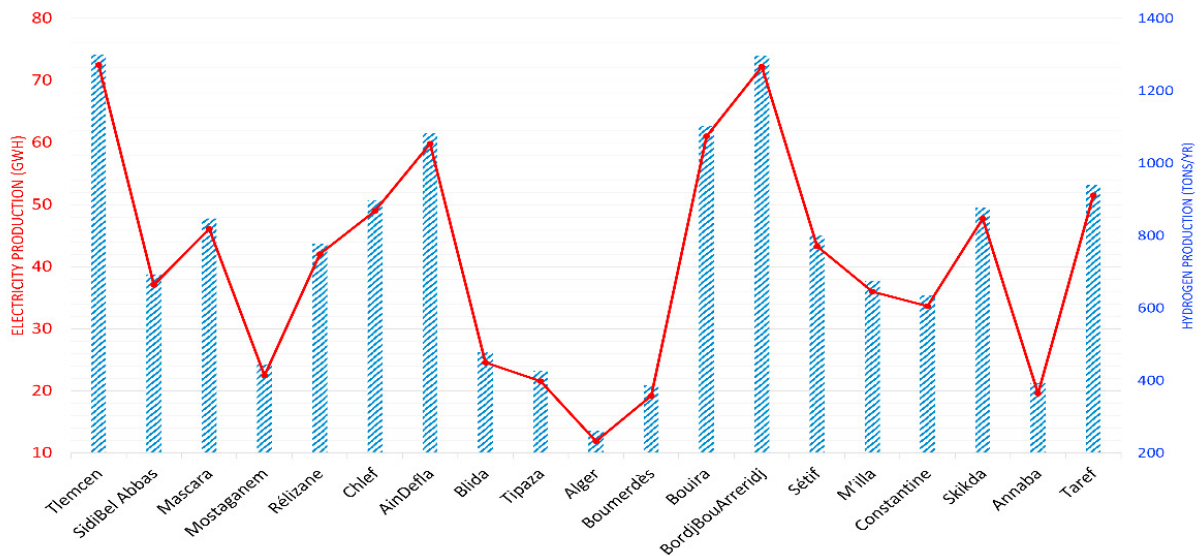


Fig.4: Electricity and hydrogen production from RSP for each province segments

In the course of this estimation, the most important issue is the hydrogen produced fulfils needs of demand in station infrastructure, in this phase we based on the classic infrastructure stations of ICEs and calculate the hydrogen supplied for every station as shown in Fig.5.

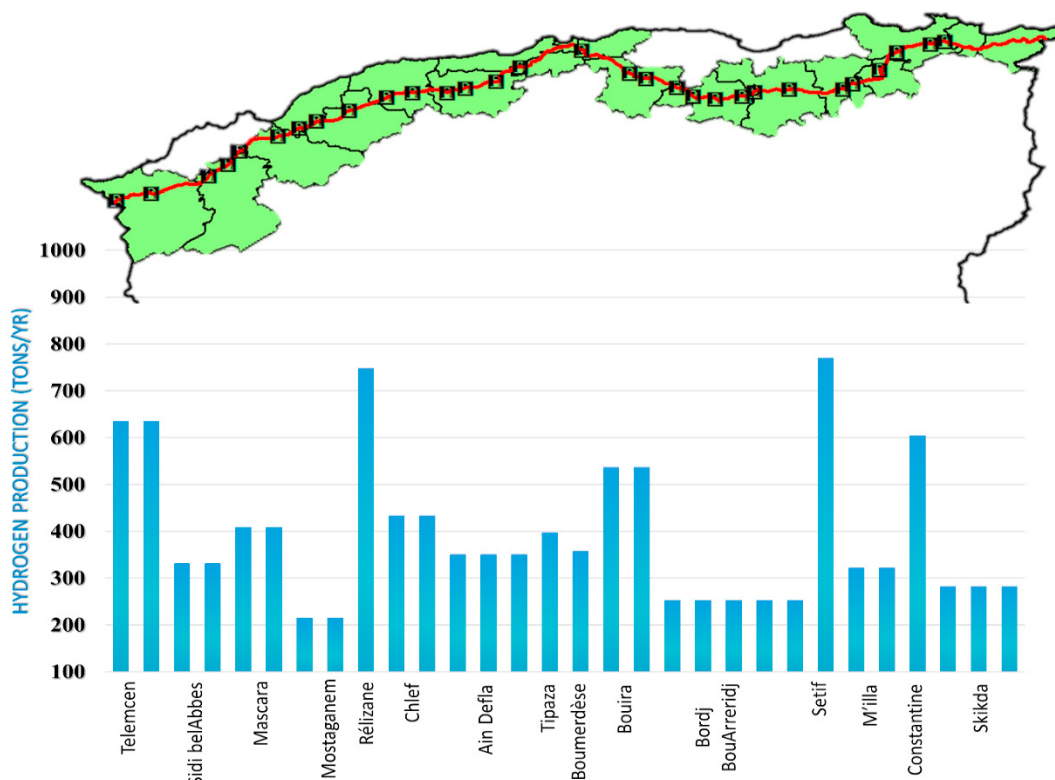


Fig.5: The amount of hydrogen supplied to conventional fuel stations.

4.1 Fuel saving and CO₂ avoided analysis

In this part of analysis, an evaluation of the CO₂ emissions elimination by using FCHEVs vehicle (1kg H₂ for each 100 km) was compared with an average gasoline vehicle (4.9 liter for each 100 km), and derived estimates of fuel savings in the same driven length, which is corresponding to drive capacity of the full hydrogen produced in the road E-O through solar road technology for one year, in **Fig .6** carbon dioxide emissions avoid for every provinces, its shown the CO₂ emissions in some parts in road E-O more than 15 Ton/year (Telemcen, Bouira, Aindefla and Taref) its due to the length of road in these provinces , the driven length isn't the only factor on the CO₂ emission they're others factors such as; the CO₂ in the fuel used by vehicle and its economy (km per kg of fuel). In other hand, the fuel saving (Gasoline / Diesel) when using a FCHEVs based on hydrogen as fuel in place of those fossil fuel, it's over than 41,000 liter in year.

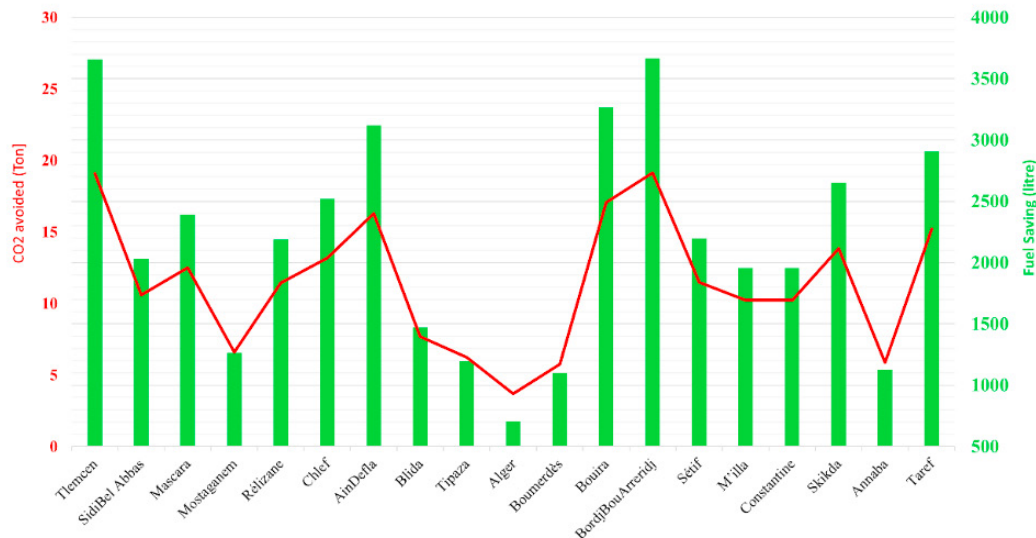


Fig.6: Amount dioxide-carbon avoid and fuel saving in RSP

5. Conclusions and Further research suggestions

This paper represents a combination between geographical data (environmental characteristics, climate, and location) and software programs that provided a scientific data which had been used in our analysis. the analysis is aim to the possibility of integrating solar roads technology into the transport sector in Algeria, therefore; this study has not only served to assess the ability to produce electricity via solar roads technology which will be used to supply power to charging stations of Plug-in-FCEVs, but it also given how it possible to produce a hydrogen by using the electricity from solar roads panels. The hydrogen produced will be used in supplying hydrogen charging stations.

As a result, the yearly electricity production is over 804 GWh of solar energy incident on road E-O. the hydrogen which can be produced using that's electricity is 13778.39 Tons per year; which can considered as a significant step towards reducing CO2 emissions by elimination a 216 Tons of CO2 emissions and result of 41.10^3 litter of fuel saving and result of 41.10^3 litter of fuel saving, by using Plug-in-FCEVs vehicle instead of internal combustible engine vehicle. Three suggestions for further research can be condensed:

- Optimization the best location of the hydrogen stations on the roads E-O.
- Study the effect of seasons on hydrogen production.
- The amount of electricity needed to feed the EVs station.

References

- [1] P. Nunes, R. Figueiredo, and M. C. Brito, "The use of parking lots to solar-charge electric vehicles," *Renew. Sustain. Energy Rev.*, vol. 66, pp. 679–693, Dec. 2016.
- [2] IEA, "CO2 emissions from fuel combustion 2016," 2017.
- [3] P. Moriarty and S. J. Wang, "Can Electric Vehicles Deliver Energy and Carbon Reductions?," *Energy Procedia*, vol. 105, pp. 2983–2988, May 2017.
- [4] J. C. González Palencia, M. Araki, and S. Shiga, "Energy consumption and CO2 emissions reduction potential of electric-drive vehicle diffusion in a road freight vehicle fleet," *Energy Procedia*, vol. 142, pp. 2936–2941, Dec. 2017.
- [5] A. Lahnaoui, C. Wulf, and D. Dalmazzone, "Building an optimal hydrogen transportation system for mobility, focus on minimizing the cost of transportation via truck," *Energy Procedia*, vol. 142, pp. 2072–2079, Dec. 2017.
- [6] F. Orecchini et al., "Energy consumption of a last generation full hybrid vehicle compared with a conventional vehicle in real drive conditions," *Energy Procedia*, vol. 148, pp. 289–296, Aug. 2018.
- [7] S. Singh et al., "Hydrogen: A sustainable fuel for future of the transport sector," *Renew. Sustain. Energy Rev.*, vol. 51, pp. 623–633, Nov. 2015.
- [8] E. S. Hanley, J. Deane, and B. Ó. Gallachóir, "The role of hydrogen in low carbon energy futures—A review of existing perspectives,"

- Renew. Sustain. Energy Rev.*, vol. 82, pp. 3027–3045, Feb. 2018.
- [9] F. Liu, F. Zhao, Z. Liu, and H. Hao, “The impact of fuel cell vehicle deployment on road transport greenhouse gas emissions: The China case,” *Int. J. Hydrogen Energy*, vol. 43, no. 50, pp. 22604–22621, Dec. 2018.
- [10] P. Moriarty and S. J. Wang, “Could automated vehicles reduce transport energy?,” *Energy Procedia*, vol. 142, pp. 2109–2113, Dec. 2017.
- [11] P. SAISIRIRAT and N. CHOLLACOOP, “A Scenario Analysis of Road Transport Sector: the Impacts of Recent Energy Efficiency Policies,” *Energy Procedia*, vol. 138, pp. 1004–1010, Oct. 2017.
- [12] Ministry of Energy, “Bilan des réalisations du secteur année 2016,” 2017.
- [13] Bp, “bp statistical review of world energy,” 2017.
- [14] Ministry of Energy, “Bilan du marché national des carburants terre 2016,” 2017.
- [15] S. F. Tie and C. W. Tan, “A review of energy sources and energy management system in electric vehicles,” *Renew. Sustain. Energy Rev.*, vol. 20, pp. 82–102, Apr. 2013.
- [16] M. F. M. Sabri, K. A. Danapalasingam, and M. F. Rahmat, “A review on hybrid electric vehicles architecture and energy management strategies,” *Renew. Sustain. Energy Rev.*, vol. 53, pp. 1433–1442, Jan. 2016.
- [17] K. W. Lee, A. J. Correia, B. D. J. Neilan, and B. Dahn, “Comparative Study of Solar Panels for Roadway Operations,” *Adv. Mater. Res.*, vol. 723, pp. 594–600, 2013.
- [18] “TNO Annual Report,” *Solaroad: sustainable energy from a SMART Road Surface*, 2014. .
- [19] Wattway, “colas.” [Online]. Available: <http://www.wattwaybycolas.com/en/>. [Accessed: 23-Apr-2018].
- [20] B. Joachim and S. Philippe, “Wattway paving the way to tomorrow’s energy.”
- [21] S. Patel, “New solar roads unveiled,” *Power*, vol. 161, no. 2, 2017.
- [22] Liangyu, “China’s solar expressway meets light of day,” 2017. [Online]. Available: http://www.xinhuanet.com/english/2017-12/28/c_136857894.htm. [Accessed: 01-Jul-2018].
- [23] “How To: Select a suitable map projection or coordinate system.” [Online]. Available: <https://www.esri.com>. [Accessed: 12-Jan-2017].
- [24] D. Mentis, S. Hermann, M. Howells, M. Welsch, and S. H. Siyal, “Assessing the technical wind energy potential in Africa a GIS-based approach,” *Renew. Energy*, vol. 83, pp. 110–125, Nov. 2015.
- [25] A. Gouareh et al., “GIS-based analysis of hydrogen production from geothermal electricity using CO2 as working fluid in Algeria,” *Int. J. Hydrogen Energy*, vol. 40, no. 44, pp. 15244–15253, Nov. 2015.
- [26] S. Dahbi, R. Aboutni, A. Aziz, N. Benazzi, M. Elhafyani, and K. Kassmi, “Optimised hydrogen production by a photovoltaic-electrolysis system DC/DC converter and water flow controller,” *Int. J. Hydrogen Energy*, vol. 41, no. 45, pp. 20858–20866, Dec. 2016.
- [27] H. Dagdougui, A. Ouammi, and R. Sacile, “A regional decision support system for onsite renewable hydrogen production from solar and wind energy sources,” *Int. J. Hydrogen Energy*, vol. 36, no. 22, pp. 14324–14334, Nov. 2011.
- [28] J. Chi and H. Yu, “Water electrolysis based on renewable energy for hydrogen production,” *Chinese J. Catal.*, vol. 39, no. 3, pp. 390–394, Mar. 2018.
- [29] S. Rahmouni, B. Negrou, N. Settou, J. Dominguez, and A. Gouareh, “Prospects of hydrogen production potential from renewable resources in Algeria,” *Int. J. Hydrogen Energy*, vol. 42, no. 2, pp. 1383–1395, Jan. 2017.
- [30] S. Rahmouni, N. Settou, B. Negrou, and A. Gouareh, “GIS-based method for future prospect of hydrogen demand in the Algerian road transport sector,” *Int. J. Hydrogen Energy*, vol. 41, no. 4, pp. 2128–2143, Jan. 2016.
- [31] ADEME, “base carbone,” 2014.
- [32] B. L. Salvi and K. A. Subramanian, “Sustainable development of road transportation sector using hydrogen energy system,” *Renew. Sustain. Energy Rev.*, vol. 51, pp. 1132–1155, Nov. 2015.
- [33] ONS, “Office National des Statistiques,” *Fleet of cars*, 2017. [Online]. Available: <http://www.ons.dz>. [Accessed: 06-Feb-2018].
- [34] TOYOTA, “Mirai Product Information,” 2016.